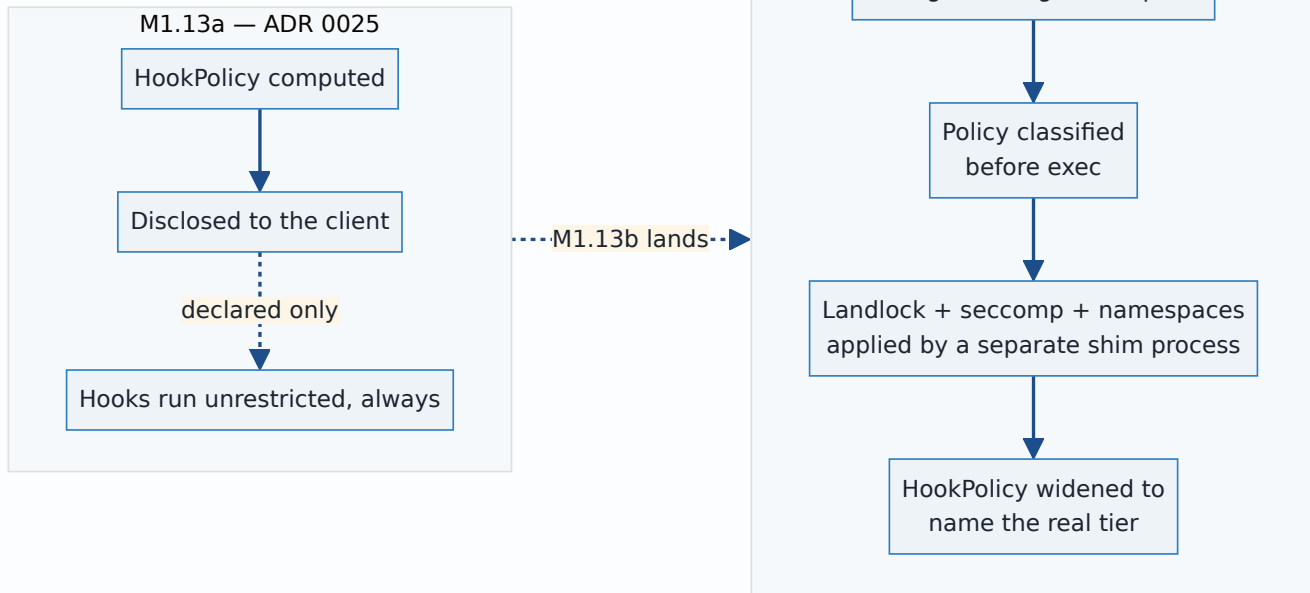


0030 — The git-process sandbox: a pure argv boundary, tiers by declared intent, and tests that prove their own premise

- **Status:** Accepted — core mechanism landed and tested; INV-15 disclosure wiring in flight (see Consequences).
- **Date:** 2026-07-30
- **Milestone / issue:** M1.13b — the git-process sandbox (#66). Written for issue #205 (Task 18, "the whole-sandbox ADR and security-model closure"), the record ADR 0027 already reserved a slot for when it warned a reader off looking for it at "ADR 0026."
- **Supersedes:** nothing directly — this is the cross-cutting record the four standalone M1.13b ADRs (0025 partially, 0027, 0028, 0029) each anticipated. **Amends:** ADR 0025, via the amendment appended to that file rather than an edit to it — `HookPolicy` widened from two variants to four, and the plan's `CapabilityAbsent` → `Blocked` mapping ADR 0029 rejected is now checked, not merely rejected in prose.
- **Related:** [0025](#) (declared-and-disclosed discipline, the pattern this whole milestone follows), [0015](#) (closed vocabularies over booleans/strings — the reasoning `Tier`, `NetworkNeed`, `HookPolicy` and `GitOperation` all reuse), [0017](#) (the same sealed-argv discipline, applied there to the browser boundary and here to the git-spawn boundary), [0027](#) (the Landlock mechanism this ADR's "layered mechanisms" section summarizes rather than repeats), [0028](#) (the network tier's accepted host-confinement gap), [0029](#) (INV-13 in full; this ADR restates only what the whole picture needs).

Context

Before M1.13b, ADR 0025 shipped the *declared* half of `SECURITY_MODEL.md:236`: a `HookPolicy` value computed and disclosed, with the explicit, load-bearing caveat that "nothing in `git_cmd.rs` or `git-vista-git` reads this value or suppresses hooks accordingly" — every git process the server spawned ran with no seccomp, no Landlock, no sandbox of any kind, hooks included. M1.13b is the enforcement half: contain what a hostile repository's own hooks can do to the host, without breaking what git legitimately needs — identity from `~/.gitconfig` (nine of twenty-four repositories on this box have no repo-local identity and depend on it), and DNS resolution for `push / fetch / clone`, which a network namespace breaks (round-4 verdict finding F3).

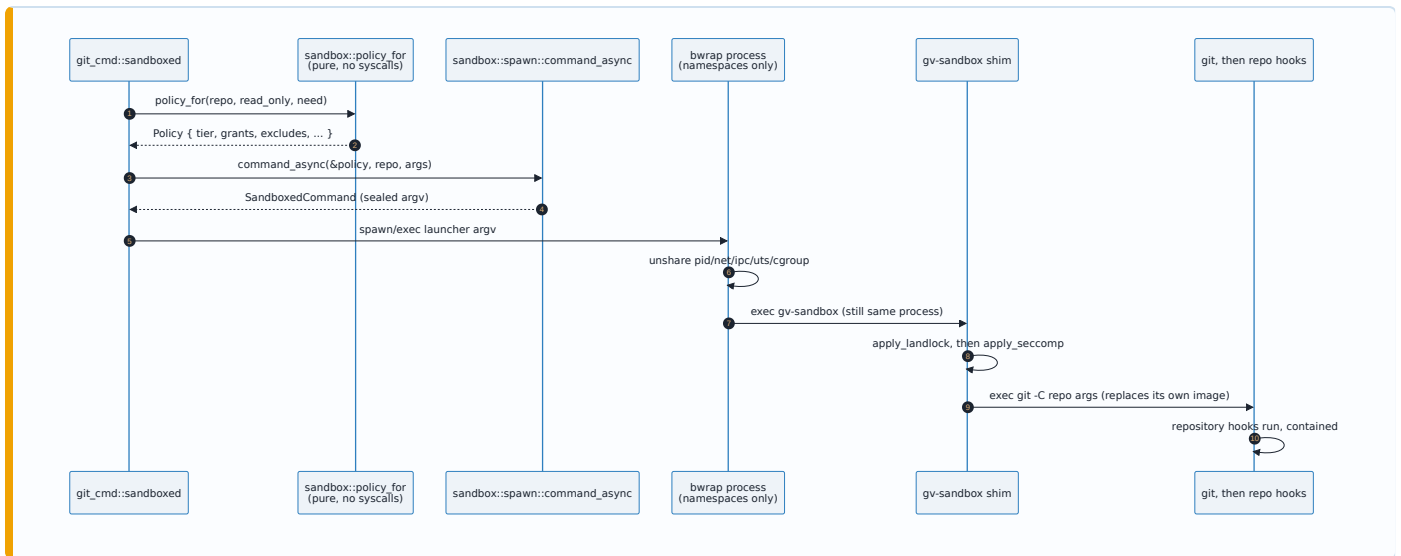


The rest of this record is organized around the properties that make the enforcement trustworthy rather than merely present: the boundary is provably the thing that runs (not a policy computed and then bypassed), trust is structurally narrow (not a convenience escape hatch), and the tests that claim containment are structurally prevented from passing for the wrong reason — this milestone's escape battery was found vacuous by audit twice before landing the contract described below.

Decision

1. The sandbox is argv — a pure policy, one impure shim

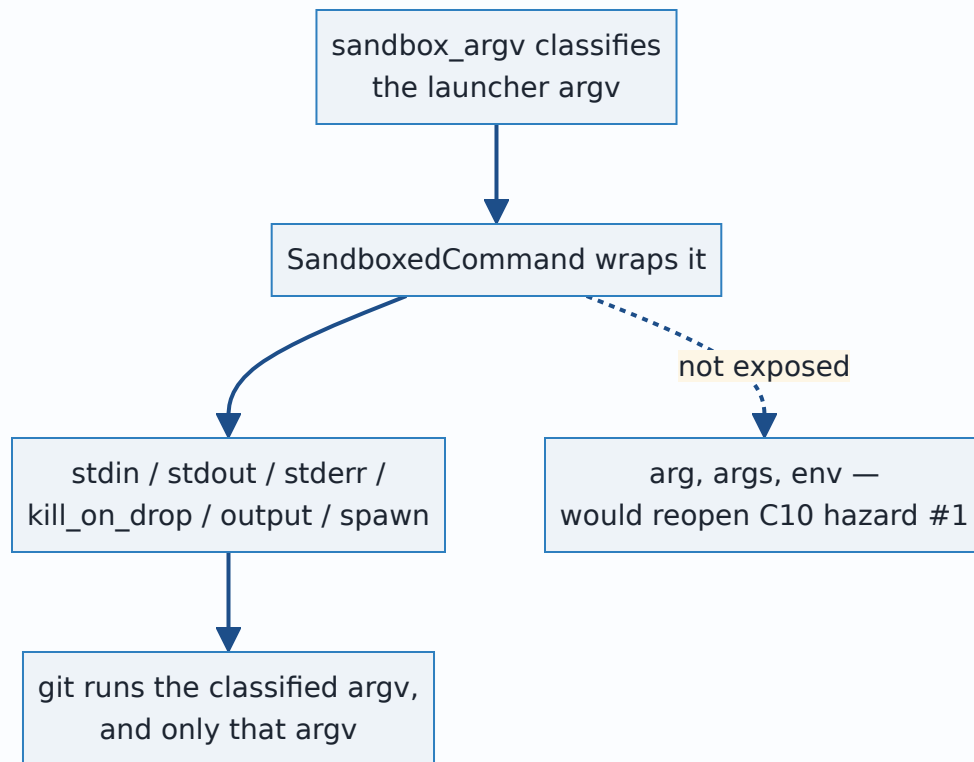
`sandbox/mod.rs` builds a `Policy` and turns it into a launcher `Vec<OsString> ( sandbox_argv )` with **no syscall, no I/O, no async** (`mod.rs:1-8`). The two genuinely impure steps — finding `bwrap` on disk, finding the `gv-sandbox` shim — are pulled into their own submodules (`bwrap`, `shim`) for exactly this reason: `sandbox_argv` stays a total function of its `Policy`. That purity is what let the sync and async spawn wrappers collapse into one function; per `spawn.rs:27-29`, "neither call style needs a `pre_exec` closure or a `block_on`, because the sandbox is *argv*: the shim applies Landlock and seccomp in its own process, after this one has already exec'd it."



The shim's own `main()` is `parse` → `validate` → `close_inherited_fds` → `apply_landlock` → `apply_seccomp` → `exec("git")` (`bin/gv-sandbox/main.rs:735-762`) — Landlock before seccomp, one process, `.exec()` never `.spawn()` , so the shim never becomes a parent (enforced by a source tripwire on that same file, `main.rs:10-13`).

2. A sealed argv — `SandboxedCommand` cannot be appended to

Purity upstream is not enough on its own: a bare `Command` returned from the chokepoint could still have `.args(...)` called on it after classification — this is exactly the shape the crate shipped before Task 5, where `sandboxed()` built an *empty*-arg `Command` and every caller appended the real subcommand afterward (`spawn.rs:61-65`). `SandboxedCommand` (`spawn.rs:77`) closes it by type: its public API is `stdin` / `stdout` / `stderr` / `kill_on_drop` / `output` / `spawn` — stdio configuration only. There is deliberately no `arg` , `args` or `env` (`spawn.rs:69-73`): `GIT_DIR` , `GIT_SSH_COMMAND` and `GIT_EXTERNAL_DIFF` redirect or execute, so an environment appended after classification is an argv change wearing a different hat.

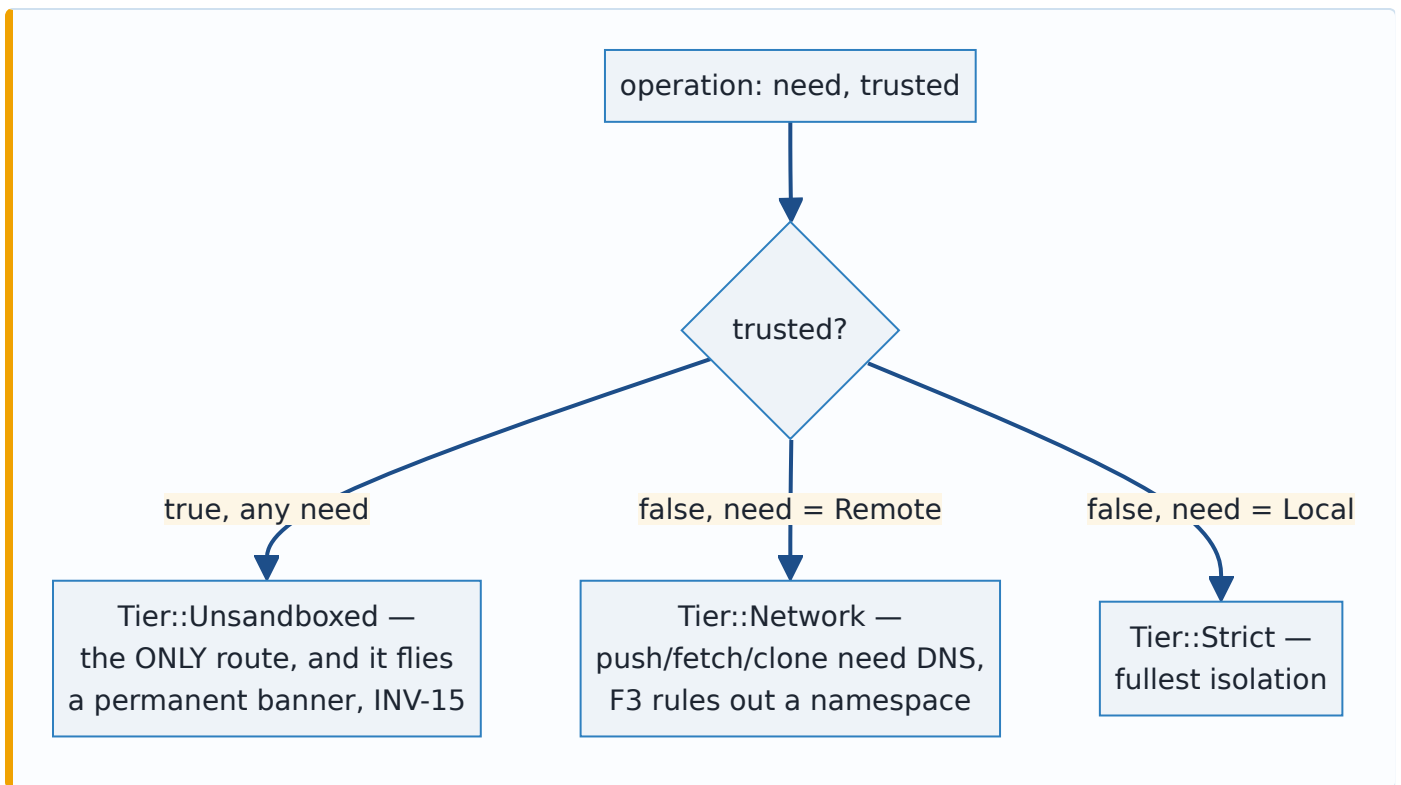


Rust has no stable negative-impl assertion, so this is enforced as a source tripwire rather than a comment: `the_sandboxed_command_exposes_no_way_to_change_what_runs` (`spawn.rs:265-295`) `include_str!` `spawn.rs` itself, isolates the `impl SandboxedCommand` block, and asserts no `pub(crate) fn` line contains `arg` , `args` or `env` — with one named, test-gated exception (`hermetic_env_for_test`) that production code cannot reach.

3. Three tiers, and `Unsandboxed` reachable only through persisted trust

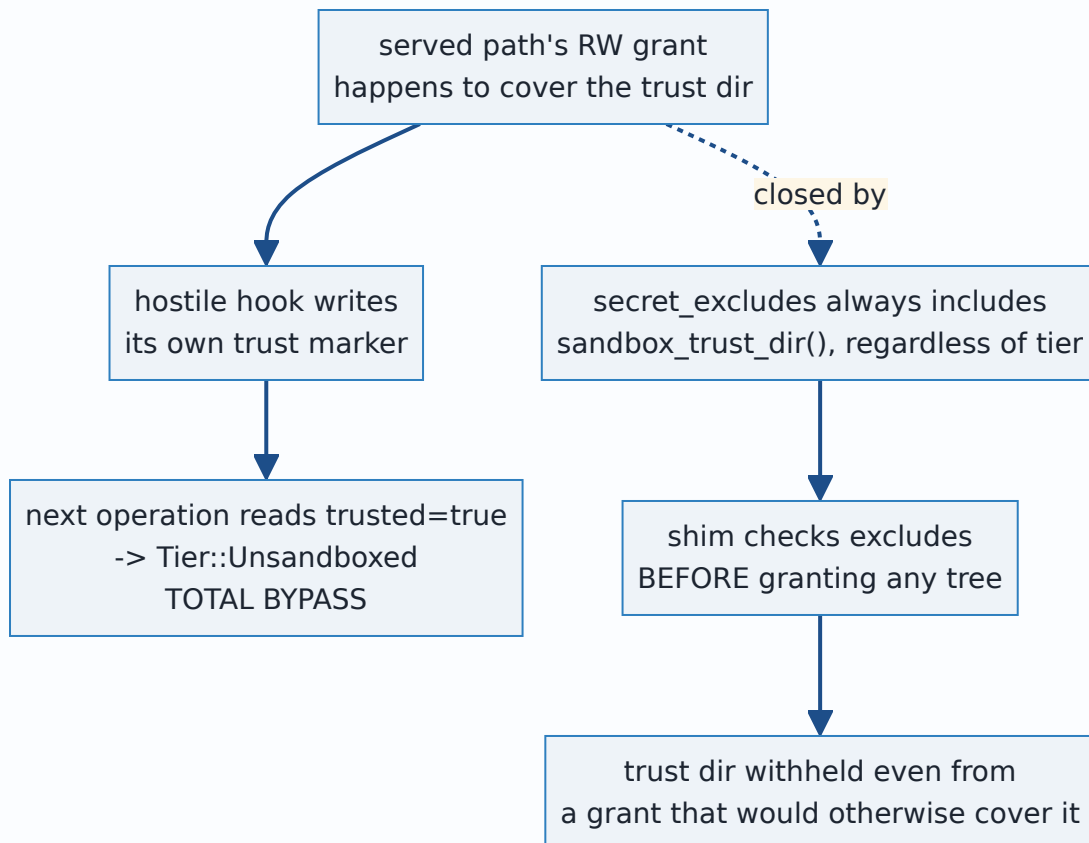
`Tier::{Strict, Network, Unsandboxed}` (`mod.rs:404-414`). Dispatch is two inputs, three outcomes, no wildcard on the untrusted side:

```
// mod.rs:781-792
pub(crate) fn tier_for(need: NetworkNeed, trusted: bool) -> Tier {
    match (trusted, need) {
        (true, _) => Tier::Unsandboxed,
        (false, NetworkNeed::Remote) => Tier::Network,
        (false, NetworkNeed::Local) => Tier::Strict,
    }
}
```



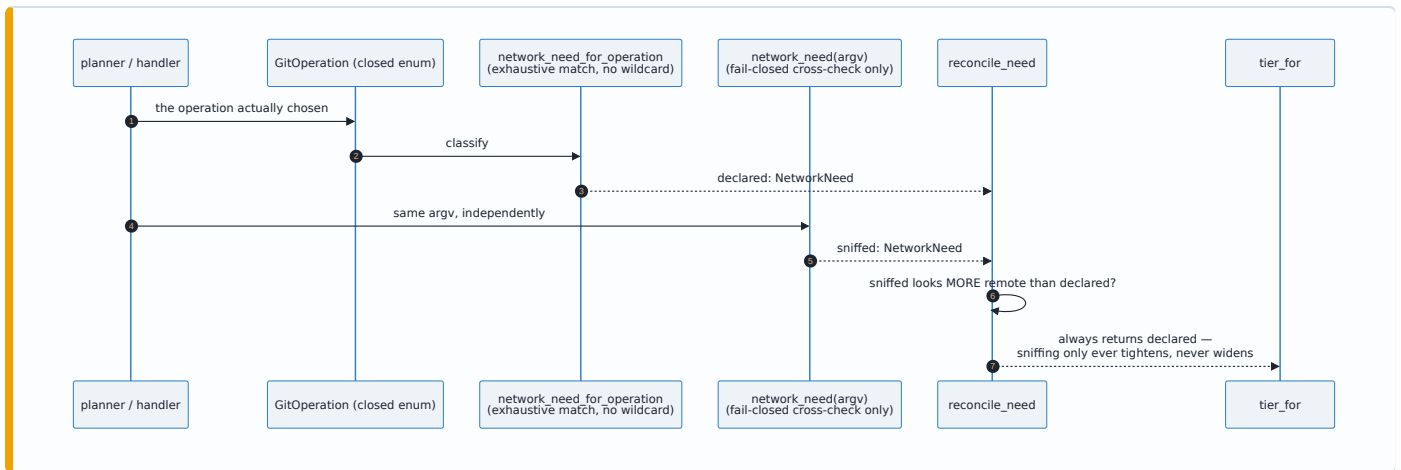
(true, _) uses a wildcard on purpose (trust is a property of the repository, not the operation); the false side has none, so a new `NetworkNeed` variant forces a new arm that cannot silently resolve to `Unsandboxed` without an edit a reviewer would see (mod.rs:758-767). `trusted` comes only from `repo_is_trusted` → `trust::is_trusted`, backed by a marker file under the server's own state directory (`state::sandbox_trust_dir()`, `$XDG_STATE_HOME/git-vista/trusted-repos` or `~/.local/state/...`) — never inside a repository, and the only writer is `trust::grant`, documented as callable "only from an explicit, authenticated operator action" (`trust.rs:76-90`). As of this writing `trust::grant` has no production (handler-reachable) caller — only test code exercises it (`sandbox/dispatch.rs:616,648`, `sandbox/compat.rs:471`) — so `Unsandboxed` is reachable by rule, not yet by any operator-facing route.

A read-only `$HOME` grant is not, by itself, what protects that marker: serving a repository whose grant happens to cover `~/.local/state` (or a served path pointing `XDG_STATE_HOME` inside itself) would otherwise let a hostile hook write its own trust marker and promote itself to `Unsandboxed` on the next operation — a total bypass, named explicitly in source as the reason the fix is a hard exclude, not the RO grant (mod.rs:984-1010). `secret_excludes` always carries `sandbox_trust_dir()` in addition to the standard secret list (mod.rs:1011-1015), and the shim's grant-building functions check excludes *before* deciding whether or how to grant a tree at all (`is_or_inside_exclude / is_ancestor_of_exclude`, `bin/gv-sandbox/main.rs:527-563`) — the one mechanism in this sandbox that outranks a grant rather than competing with it, tested directly by forging exactly that scenario (`trust.rs:177-232`).



4. Declared intent outranks argv sniffing

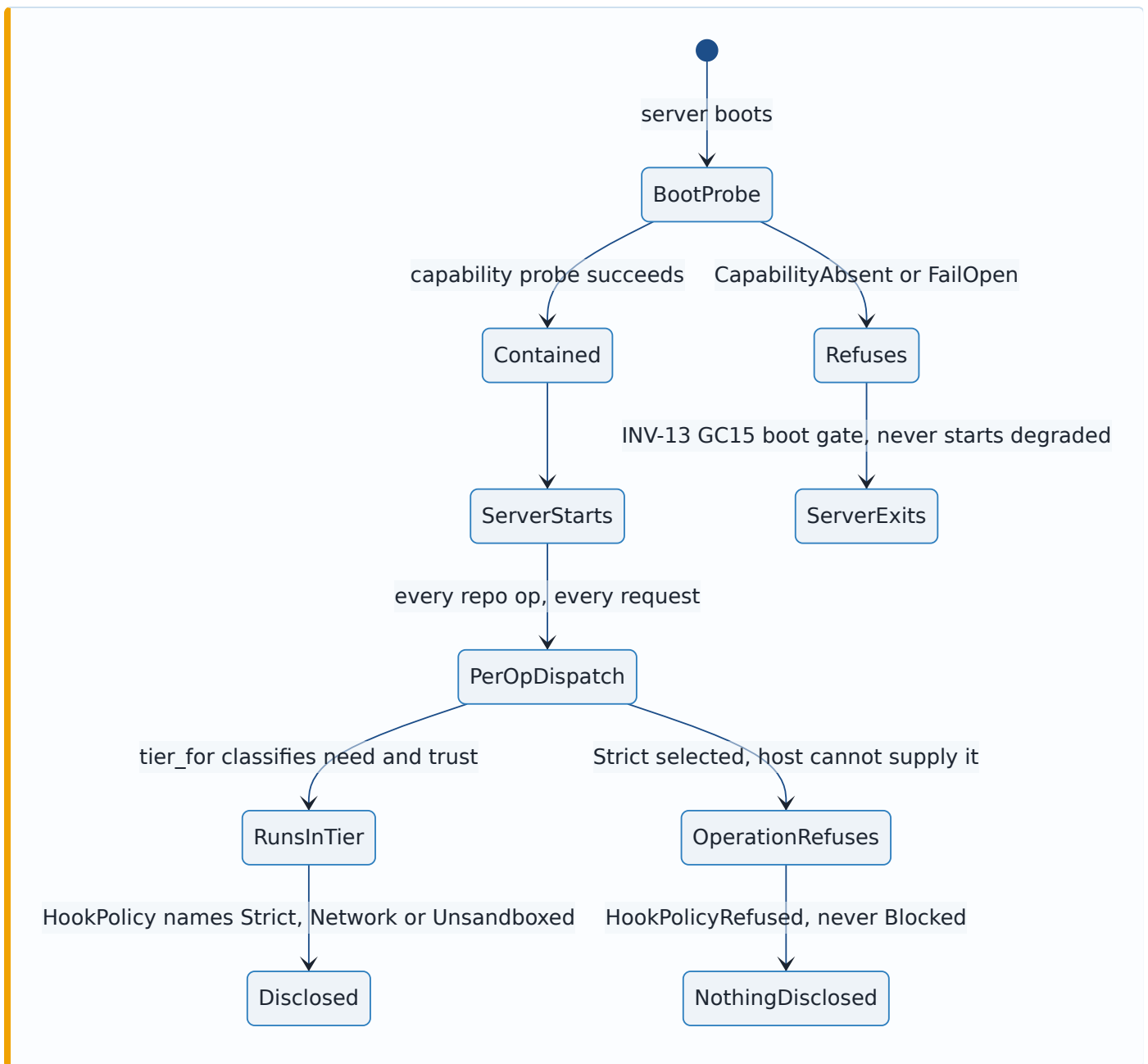
`GitOperation` is a closed, data-carrying enum in `git-vista-protocol` (`plan.rs:143-153`, ADR 0015's vocabulary reused). `network_need_for_operation` is an exhaustive match over it with no wildcard arm (`mod.rs:593-621`) — a new operation variant fails the build until someone states what network it needs, rather than silently inheriting a default. The argv itself is also inspected (`network_need`, `mod.rs:539-563`), but only as a fail-closed cross-check: the reconciler `reconcile_need` always returns the *declared* value, only ever logging/panicking-in-debug when the argv looks more remote than declared (`mod.rs:661-679`, `mod.rs:657-660`: "the argv can only ever *complain*, never decide").



5. INV-13 (already ADR 0029) and INV-15 disclosure, as they sit today

Restated only for the whole picture — ADR 0029 is the detailed record. `policy_for` refuses *before* building any grant when `Strict` is selected and the host cannot supply it (`strict_launcher` , `mod.rs:704-726` , `ShimError::StrictUnavailable`), never degrading to `Network` and never running with hooks suppressed. The disclosure seam makes the same refusal honest at the wire:

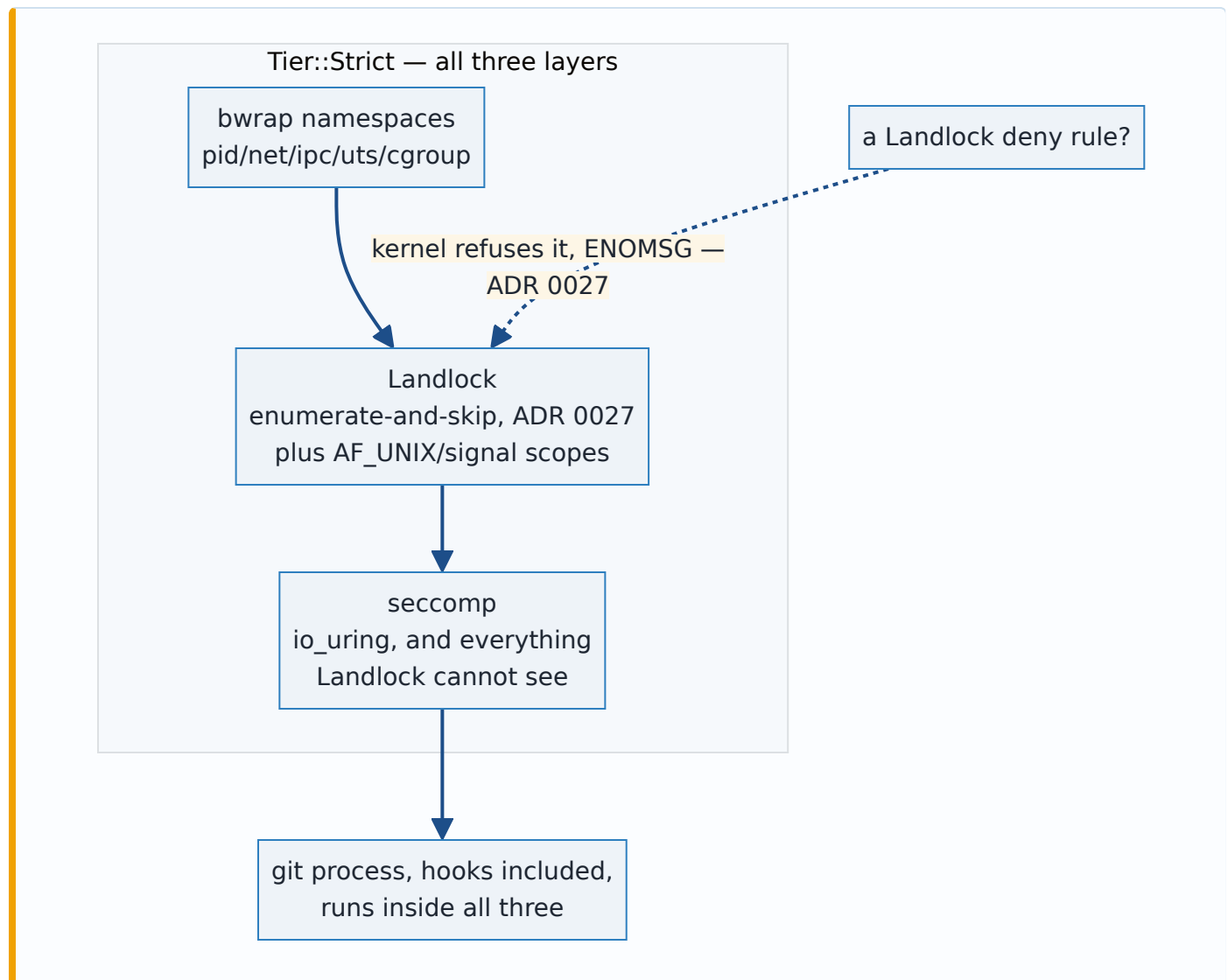
`hook_policy_for_repo / hook_policy_for_trusted_repo` maps `ProbeVerdict::CapabilityAbsent / FailOpen` to `Err(HookPolicyRefused::...)` , never to `HookPolicy::Blocked` — the exact mapping the M1.13b plan proposed and ADR 0029 rejected by name. This is checked, not merely documented: `capability_absent_refuses_and_never_becomes_blocked` (`hook_policy.rs:249-264`) asserts both `is_err()` and `assert_ne!(got.ok(), Some(HookPolicy::Blocked))` .



`HookPolicy` itself widened from ADR 0025's two variants to four (`git-vista-protocol/src/dto.rs:141-185`): `Strict`, `Network`, `Unsandboxed` name the three tiers directly; `Blocked` is wire-only ("hooks are not known to be running" — `HookPolicy::default()`, `dto.rs:207-220`) with **no production policy constructor emitting it**, checked by the escape contract's own R8 scan. `requires_banner()` is `!matches!(self, Strict)` (`dto.rs:202-204`) — INV-15's banner marks everything that is not the fullest isolation, `Blocked` included, since "your hooks silently did not run" is a surprise a user must be told about as much as "your hooks ran unsandboxed." ADR 0025's old wire strings (`allow / restricted`) still deserialize via `#[serde(alias)]`, a wire-compatibility guarantee kept deliberately separate from the Rust-level transition constants, which have since been deleted now that every call site spells a tier name directly.

6. Layered mechanisms, and why denial is expressed by omission

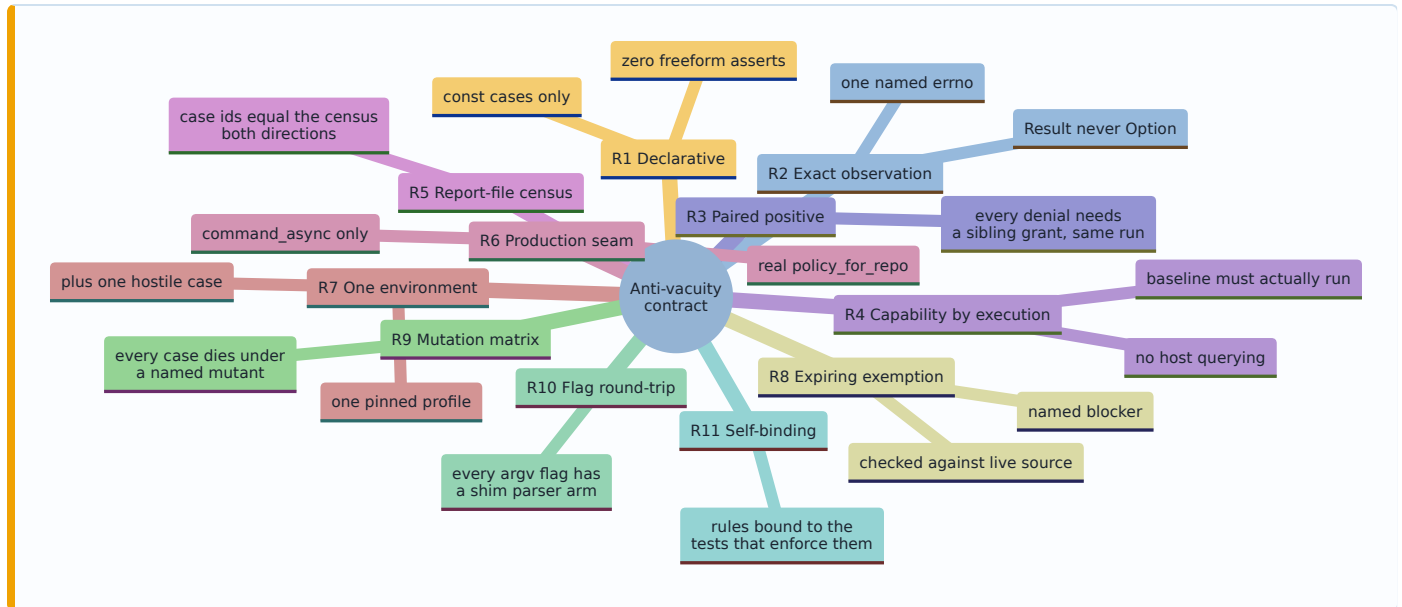
Strict composes three independent layers: bwrap namespaces (pid/net/ipc/uts/ cgroup), Landlock (filesystem and, since ABI 6, AF_UNIX /signal scopes), and seccomp (syscall classes Landlock cannot mediate — `io_uring`, which submits path opens from kernel space and bypasses Landlock's open-time checks entirely). Landlock is deny-by-default with no rule shape that subtracts from an already-granted tree (ADR 0027, measured: a `path_beneath` rule with `allowed_access = 0` is refused by the kernel with `ENOMSG`; a nested lower-privilege rule is simply inert). Denial is therefore expressed only by **never granting** — enumerate the tree, add one rule per entry, skip anything in the exclude set — which is also the mechanism Decision 3 above leans on to protect the trust marker.



Denial-by-omission is why a path that is legitimately granted through an alias — a symlink at an enumerated depth, a hard link to a secret — can silently void an exclusion (ADR 0027's inode-identity fix); the AF_UNIX seccomp gap (`socket` / `socketpair` unmediated, plan claimed otherwise in two places, enforced in neither) was found and closed the same way, strict tier only (`seccomp_filter.rs`, INV-4).

7. The anti-vacuity contract — R1 through R11

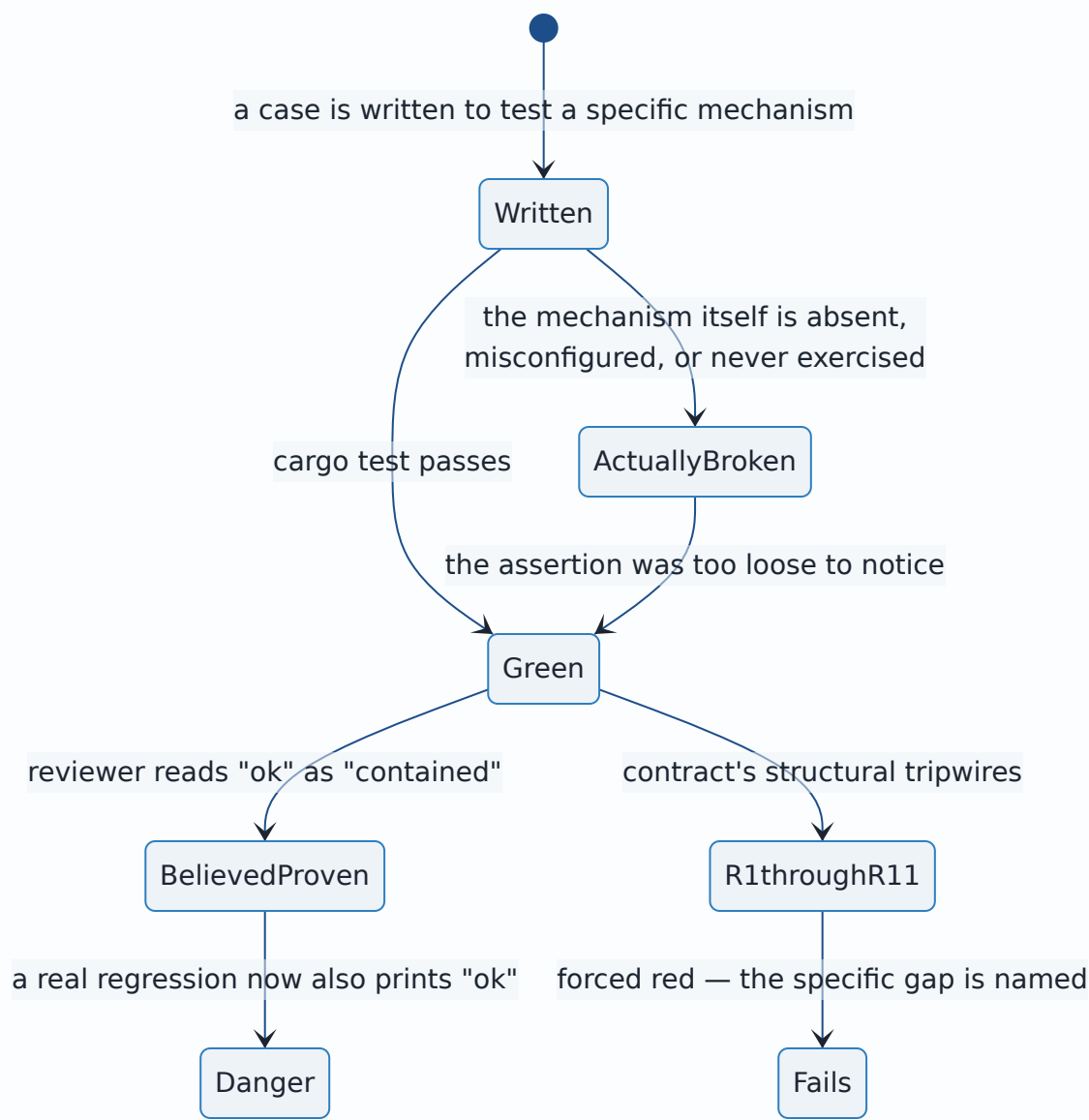
The escape battery failed audit twice before this landed: once found vacuous (C8), rewritten, and the rewrite's `io_uring` case found vacuous again by a second audit (C11: "0 PROVES, 4 VACUOUS, 1 UNCERTAIN") — the defect had "merely moved from the inside assertion to the baseline gate," not been removed. `docs/sandbox/escape-battery-anti-vacuity-contract.md` responds with a checkable standard instead of a third rewrite guided by care: eleven rules, each either a **source tripwire** (fails the build) or a **CI step** (fails the gating job on an artifact both a real run and a broken one must produce identically).



R3 is the rule that closes the sharpest hole: because Landlock exclusion is implemented as omission-during-enumeration, a denial on an excluded path and a denial on a path that was simply never granted are the *same kernel event* — a test asserting only the denial cannot tell them apart, so every case must also assert a sibling grant succeeds under the same policy, same run. R9's mutation matrix is what turns "the mechanism is denied" into "*this specific* case would notice if the mechanism broke": M8 removes only the strict tier's `AF_UNIX` seccomp rules, leaving the rest of the filter installed, so only a case that depends on that exact rule — not "some seccomp rule" — dies under it.

8. Six tests that were green while proving nothing

The pattern the contract exists to stop, concretely — verified against current source, with one correction to the popular retelling noted:



1. **TIME-WAIT residue voided a case.** `strict_tcp_bind_denied`'s baseline bound a fixed port; a TIME-WAIT socket left by an unrelated earlier test made the bind fail with `EADDRINUSE`, the case reported `CapabilityAbsent`, and `cargo test` printed `ok` while asserting nothing about containment (`escape_contract.rs:784-791`).
2. *(Reported elsewhere as "a module never compiled" — could not be confirmed against this repository's escape-battery material; see the Report at the end of this ADR.)*
3. **A `chmod` test almost proved its own grant permits writes.** An earlier draft of `landlock_does_not_mediate_chmod` pushed its own target directory into `policy.rw_trees` and then observed `chmod` succeeding there — which a read-write grant is supposed to allow regardless of the mechanism under test (`documented_gaps.rs:88-93`, F-NEW-3). Caught before landing; the surviving test asserts the target lies under no grant at all.

4. **A census silently drifted from the source it was meant to police.**

`high_bit_af_unix_denied` and `high_bit_io_uring_denied` landed, both green, both writing report records the gating job would have diffed against a census that did not know about them — the gate would have gone red on the next CI run for a reason indistinguishable from a real regression (`escape_contract.rs:1171-1177`).

5. **A push test passed over a literal IP while DNS inside the sandbox was dead.** Before the network tier's DNS-resolver grant existed, `git ls-remote https://github.com/...` failed to resolve any host from inside it — but the push test in this crate used `git://127.0.0.1:9418`, which needs no resolver, and passed regardless: a green test over a broken feature (`mod.rs:275-283`).

6. **An AF_UNIX denial was claimed and enforced in neither of the places that claimed it.** Correction to the common retelling: it is one document making the claim twice (the M1.13b plan's Architecture section and its Task 4 text), not two independent documents — `seccomp_filter.rs` had no `SYS_socket / SYS_socketpair` rule at all. Measured inside the full strict stack: `socket(AF_UNIX, ...)` succeeded, byte-identical to the bare host. Closed 2026-07-29; the escape census now carries `af_unix_socket_denied / af_unix_socketpair_denied`, `dies_under: [M1, M8]`.

Alternatives considered

- **Represent the trust marker as anything other than a path outside every repository, hard-excluded.** A permission-bit or uid check on the marker file was not built; the anti-forgery property is entirely path-and-exclude-based (never writable by a hook, given exclude precedence over grant). Sufficient because the marker's only writer is `trust::grant`, itself unreachable from request handling.
- **Query-based capability probing in the escape battery (R4).** `shim_cli::strict_available()` — "does a bwrap binary merely exist" — said yes on stock Ubuntu 24.04 while `capabilities::strict_available()` (the real check: Landlock floor **and** bwrap **and** usable users) said no; bwrap then failed to launch and the loose check's case still printed `ok`. Rejected: execute the baseline and observe it, or record `CapabilityAbsent` — never ask the host a question the process is about to contradict.
- **A `--self-probe shim mode (R10)`, to remove `None -means-pass ambiguity`.** Rejected: a self-probe never crosses `execve`, and `execve` is precisely where production always goes and where a policy can be lost — a perfect self-probe truth table is evidence about a process configuration production never runs. R2's `Result`-typed parser plus a BEGIN/END nonce removes the same ambiguity with no shim change. `probe_argv` and its two tests were deleted rather than kept as a second, dead-on-contact route.
- **Decide test severity inside Rust (panic or early return on a missing capability).** The contract's own design (`escape-battery-anti-vacuity-contract.md`, "Skip policy") argues the opposite explicitly: "the tests never decide severity; the job does," by diffing a report-file census outside Rust, because libtest swallows stdout/stderr on a *passing* test, so a `SKIPPED` string never reaches a CI log without `--nocapture` — a gate built on that string is silently identical whether or not a skip occurred. Noted as a live tension in Consequences below:

current source has since added a `panic!` on `CapabilityAbsent` in `run_case` for a specific reason (the TIME-WAIT incident), which the contract document's own prose has not been reconciled against.

- **Degrade Strict to Network, or degrade-and-block-hooks, when Strict is unavailable.** Both rejected by ADR 0029; not re-litigated here.
- **A Landlock deny rule for excluded paths, and per-host network confinement for the network tier.** Both rejected by ADR 0027 and ADR 0028 respectively, for reasons specific to each; not re-litigated here.

Consequences

- **Every local git operation the server spawns for an untrusted repository now runs inside a tier it never ran in before M1.13a/b — Strict or Network, never bare.** Verified directly: `git_cmd.rs`'s `sandboxed()` (the crate's sole production spawn seam) calls `sandbox::policy_for`, the real dispatcher, not a hardcoded stand-in (`git_cmd.rs:207-215`). `Unsandboxed` exists and is reachable by rule, but `trust::grant` has no handler wired to it yet — a real security lever with the safety catch still on, not a decorative one.
- **Extra latency on every sandboxed spawn, measured.** The strict-tier launcher (`bwrap` + `Landlock` + `seccomp` + network namespace) costs an extra 17-24 ms per git process spawned versus `unsandboxed` (`design-docs/2026-07-29-m1.13b-escape-battery-25b.md:209-211`) — roughly 2.5-3.5x a ~7 ms `unsandboxed` baseline. Not measured anywhere in this repository: `bwrap`'s cost under the streaming/high-frequency call sites in `git_cmd.rs`, or `Strict` running a real `git log / cat-file` workload rather than the `git add` the escape suite exercises.
- **The escape battery, run honestly, is not free.** R9's mutation matrix rebuilds two crates once per mutant against the full battery — the contract's own text calls this an "honest cost" and binds it to PRs touching `sandbox/**` plus nightly, not every push (`escape-battery-anti-vacuity-contract.md`, R9). No document in this repo states a wall-clock duration for a full run; that remains unmeasured.
- **CI must relax a kernel control to run the battery at all (D6, Option A).** GitHub-hosted runners ship `kernel.apparmor_restrict_unprivileged_userns=1`, which blocks the `unshare(CLONE_NEWUSER)` `bwrap` needs. The accepted fix — a CI preflight step, `sudo sysctl -w kernel.apparmor_restrict_unprivileged_userns=0`, failing the job loudly rather than degrading if the write fails (`design-docs/2026-07-29-d6-sandbox-ci-preflight-decision.md`) — is justified specifically because GitHub-hosted runners are single-job, ephemeral VMs with no "leaves the box weaker" concern. It is landed in `.github/workflows/ci.yml`.
- **INV-15 disclosure (issue #202/Task 16) — recorded as observed, not as closed.** The wire contract is settled and committed at this ADR's HEAD: `HookPolicy`'s four variants and `RepositoryDescriptor.hook_policy: Option<HookPolicy>` both exist (`git-vista-protocol/src/dto.rs`). The *wiring* that makes a repository's real policy reach that field, and that replaces the stale `via_lan → Restricted/Allow` session-level mapping ADR 0025 introduced as a stand-in, was present in the working tree while this ADR was being written but **not yet committed** — `catalog.rs`, `handlers/session.rs`, `security.rs`, and two frontend files were locally modified, uncommitted, at the moment of verification. Issue #202's last recorded

comment named three blockers (no `RepositoryDescriptor.hook_policy`, a LAN session silencing the banner unconditionally, a plan citation to a non-existent `state::sandbox_verdict()`); source at that same moment shows fixes for all three in progress. This ADR does not assert #202 closed — only that the gap it named is not, as of this writing, an open design question, and may already be landed by the time this is read.

- **The anti-vacuity contract's own tripwires have rotted once, after landing.** R8's exemption-expiry check grepped `policy_for_repo`'s body for literal `Tier::Network / HookMode::Run`; when Task 8 removed that hard-code, the tokens survived by moving into `#[cfg(test)]` -only code, so the grep kept passing while the condition it stood for no longer existed (`escape_contract.rs:1319-1340`). The same disease this whole contract exists to catch, recurring inside the machinery built to catch it — this needs ongoing vigilance, not a one-time fix, and is tracked as its own issue (#206).
- **The skip-policy design and the shipped code disagree on where severity lives.** The contract document states plainly that "the tests never decide severity; the job does" (never a panic, never an early return). Current `run_case` panics on `Outcome::CapabilityAbsent`, added after the document was signed, in direct response to the TIME-WAIT incident (case 1 above). The hardening is reasonable on its own terms; the document has not been updated to match, so a reader of the markdown contract alone would currently be wrong about this one point.
- **The gaps ADR 0027 and ADR 0028 already accepted are unchanged by this ADR.** The network tier still confines ports, not hosts; a compromised hook in the network tier can still exfiltrate to an arbitrary host over a permitted port. Both remain documented, not fixed, by design.

Where this is implemented

- `crates/git-vista-server/src/sandbox/mod.rs` — `Tier`, `NetworkNeed`, `tier_for`, `policy_for`, `policy_for_repo`, `policy_for_clone`, `network_need_for_operation`, `network_need`, `reconcile_need`, `DEFAULT_SECRET_EXCLUDES`, `DEFAULT_GIT_PORTS`, `sandbox_argv`.
- `crates/git-vista-server/src/sandbox/spawn.rs` — `SandboxedCommand`, `command_async`, the source tripwire pinning its sealed API.
- `crates/git-vista-server/src/sandbox/trust.rs` — `grant`, `is_trusted`, the outside-every-repository marker location.
- `crates/git-vista-server/src/sandbox/capabilities.rs`, `crates/git-vista-server/src/sandbox/probe.rs` — the factual capability probe and the boot gate (`run_at_startup`, `boot_verdict`).
- `crates/git-vista-server/src/sandbox/hook_policy.rs` — `hook_policy_for_repo`, `hook_policy_for_trusted_repo`, `HookPolicyRefused`.
- `crates/git-vista-server/src/bin/gv-sandbox/main.rs`, `.../seccomp_filter.rs` — the shim: argv parsing/validation, Landlock enumerate-and-skip (`enumerate`, `is_or_inside_exclude`, `is_ancestor_of_exclude`), seccomp filter construction, the final `exec`.
- `crates/git-vista-protocol/src/dto.rs` — `HookPolicy` (four variants), `HookPolicy::requires_banner / default`, `RepositoryDescriptor.hook_policy`.

- `crates/git-vista-server/src/git_cmd.rs` — `sandboxed()` , the crate's one production spawn seam, calling `sandbox::policy_for` directly.
 - `crates/git-vista-server/src/catalog.rs` , `crates/git-vista-server/src/handlers/session.rs` — INV-15 disclosure wiring (`disclosed_hook_policy` , `session_hook_policy_for`); **in progress, uncommitted at this ADR's writing** — see Consequences.
 - `crates/git-vista-server/src/sandbox/escape_contract.rs` , `.../escape_suite.rs` , `.../hook_mode_suite.rs` , `.../documented_gaps.rs` — the escape battery and its anti-vacuity harness.
 - `docs/sandbox/escape-battery-anti-vacuity-contract.md` , `docs/sandbox/escape-census.txt` — the R1-R11 contract and its report-file census.
 - `.github/workflows/ci.yml` — the D6 Option A preflight (kernel sysctl unclamp) and the gating `sandbox` job.
 - `design-docs/2026-07-29-d6-sandbox-ci-preflight-decision.md` , `design-docs/2026-07-29-m1.13b-escape-battery-25b.md` — decision record and measurement for the two Consequences above; both untracked design-doc scratch, not the durable record (this ADR, and ADRs 0025/0027/0028/0029, are).
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Signed: thomas2025 · 2026-07-30T17:58:03-04:00